Ram Launch Initiative CDR

16 January, 2025

Colorado State University

Presentation Agenda

- Vehicle Design Review
- Payload Design Review
- Project Plan



Figure 1 – RLI's Subscale Flight 2 (Dec. 2024)

Vehicle Design Review

Launch Vehicle Dimensions

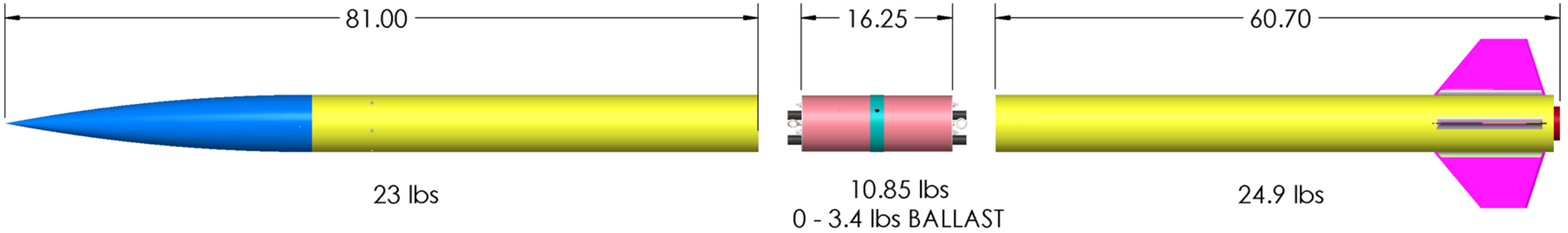


Figure 2 – Full-scale OpenRocket Model

*Masses Include Recovery Components

Key Design Features

- Sand-ballast mass based on varying wind conditions
  - Able to maintain ± 50 ft. within predicted apogee
- Payload retained with redundant bulkheads

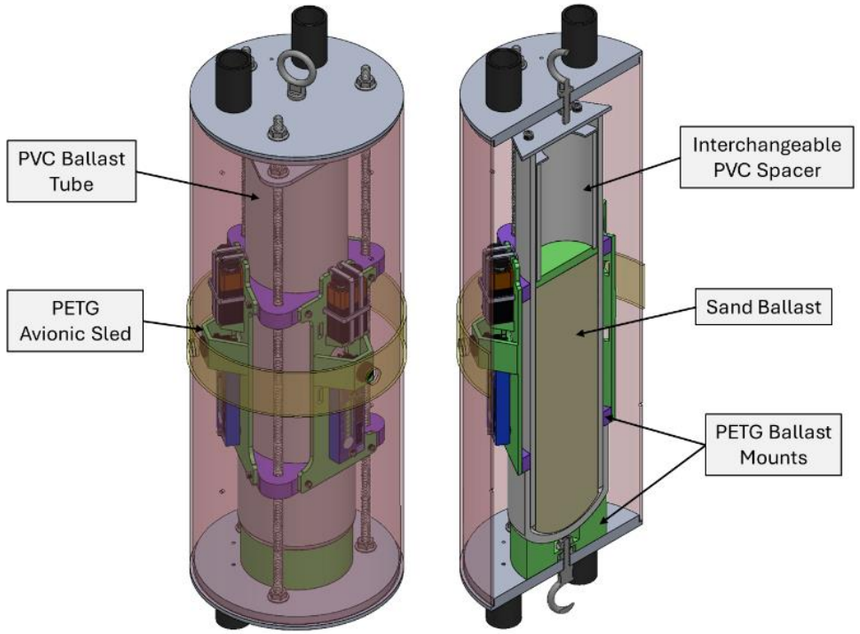


Figure 3 – Adjustable Sand Ballasting System

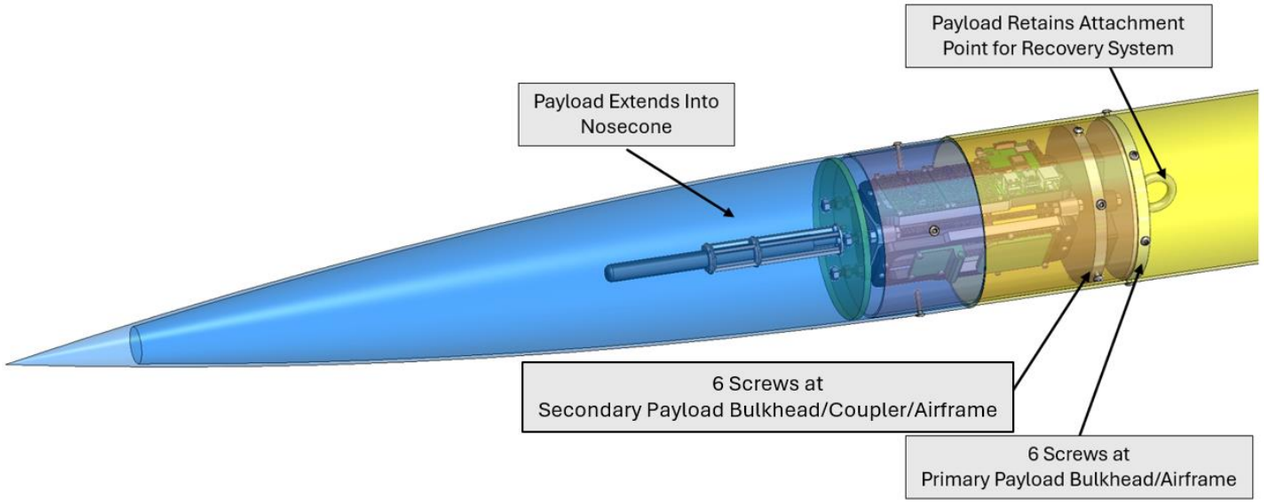


Figure 4 – Payload Integration to Vehicle

Recovery Overview

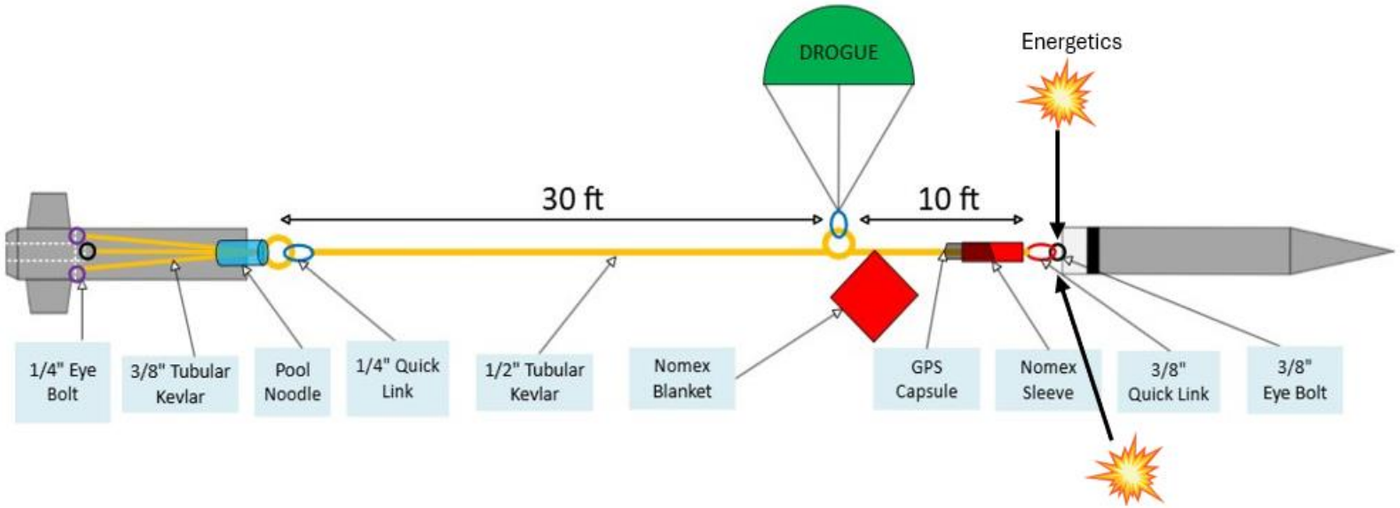


Figure 5 - Drogue Recovery Section

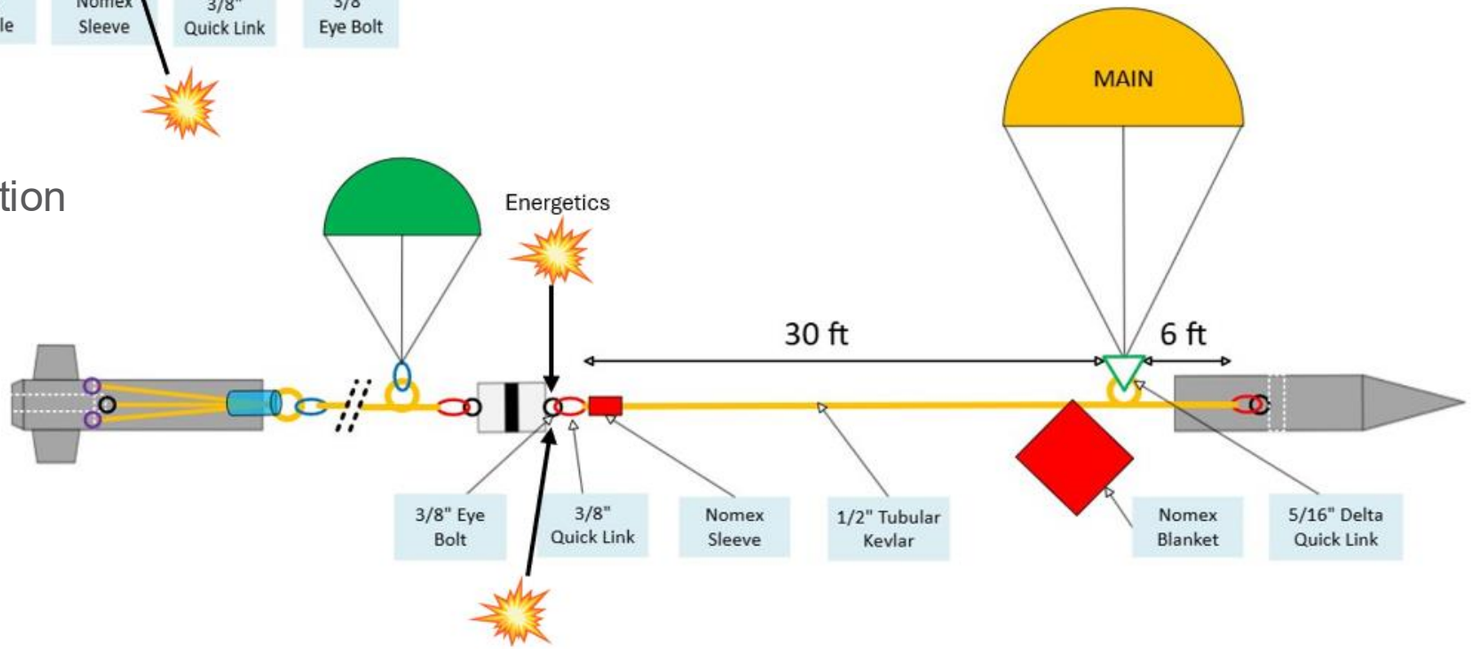


Figure 6 - Main Recovery Section

Final Motor Choices

- Final motor choice: L2200G
- Backup motor: L1420R

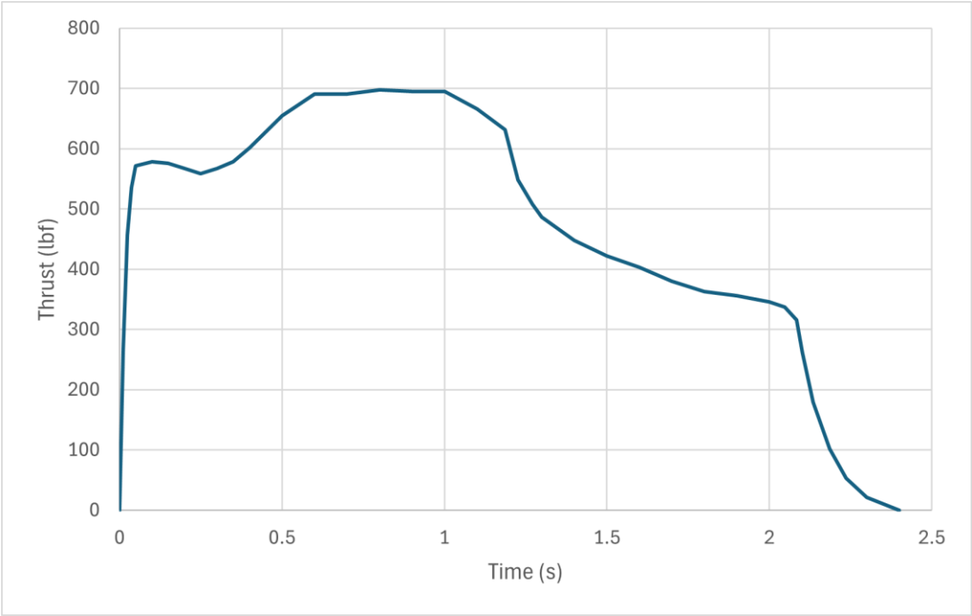


Figure 7 – L2200G Thrust Curve

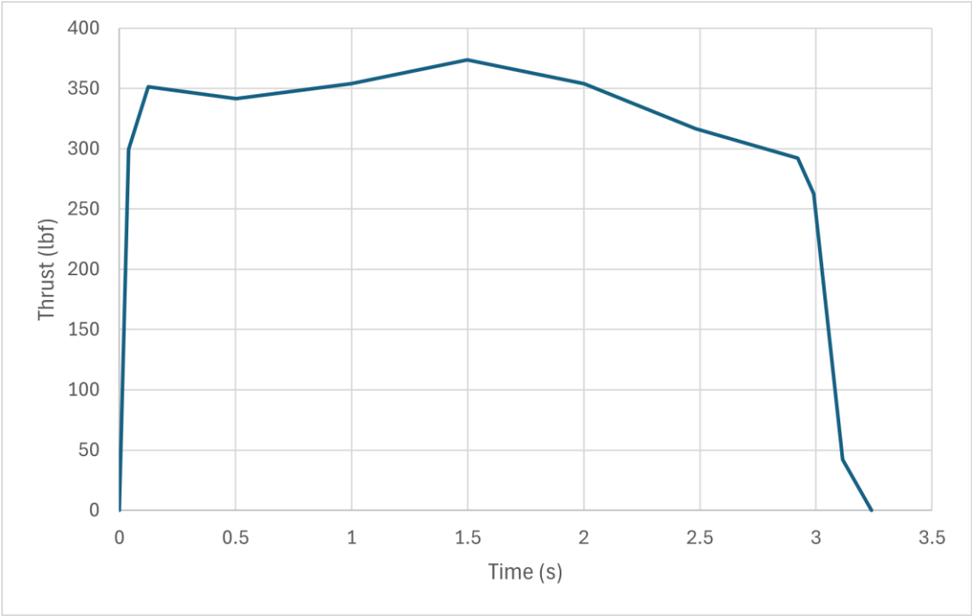


Figure 8 – L1420R Thrust Curve

Official Target Altitude

- 4,600 ft.

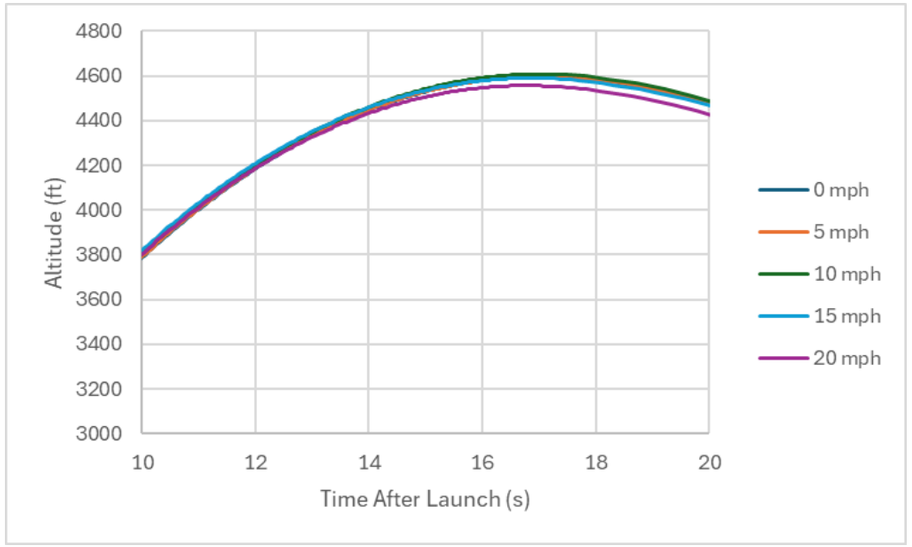


Figure 9 – Altitude Precision With Adjustable Ballast

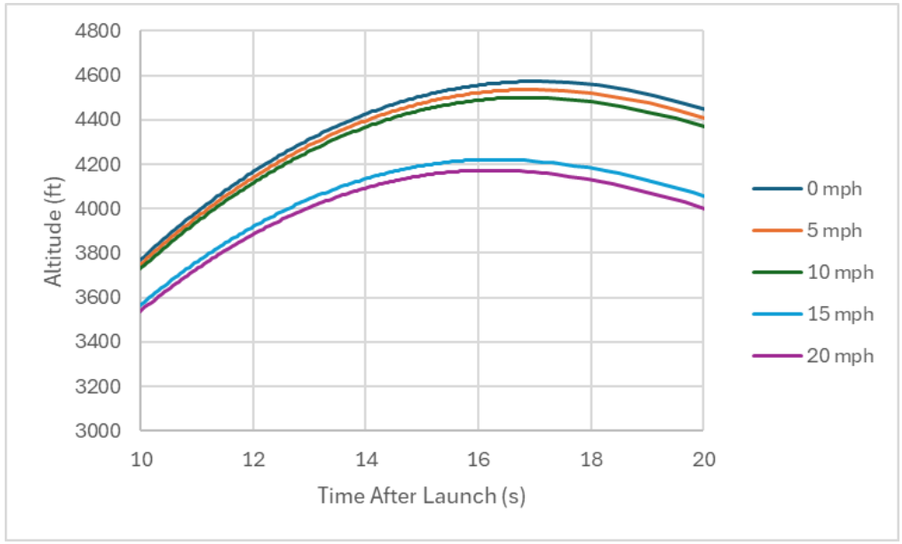


Figure 10 – Altitude Precision Without Adjustable Ballast

Wind Speed (mph)	Launch Rail Angle (deg)	Ballast Mass (lbm) *
0 mph	5	3.2
5 mph	5	2.8 lb
10 mph	5	2.2 lb
15 mph	10	0.1 lb
20 mph	10	0.0 lb

*Mass of ballast required to maintain the target apogee

Vehicle Flight Stability Margin

- 4.6 static stability caliber
- 3.5 stability caliber at rail exit (5 mph winds)
- 2.0 stability caliber at rail exit (20 mph winds)

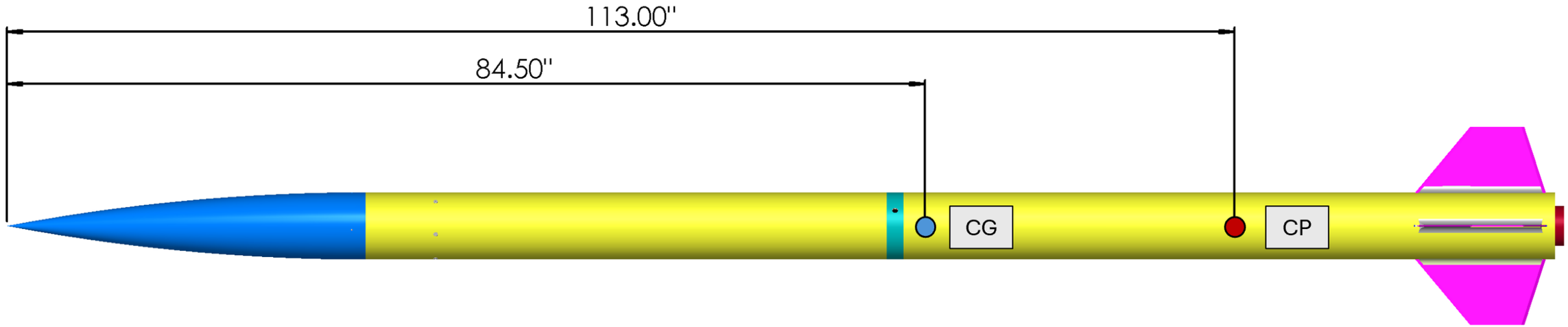


Figure 11 – Vehicle CG and CP

Thrust-to-Weight & Rail Exit Velocity

- Thrust-to-weight: 9.5 (maximum ballast)
- Rail exit velocity: 82.6 ft/s (maximum ballast)

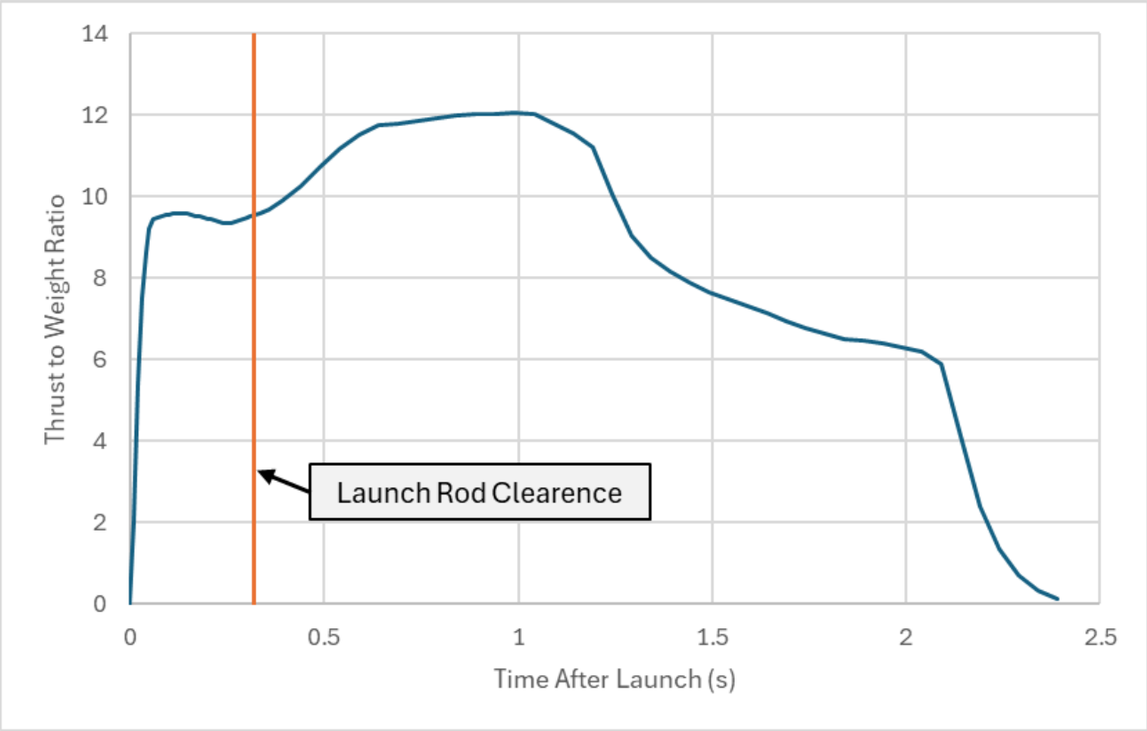


Figure 12 – Thrust-to-Weight Ratio During Flight

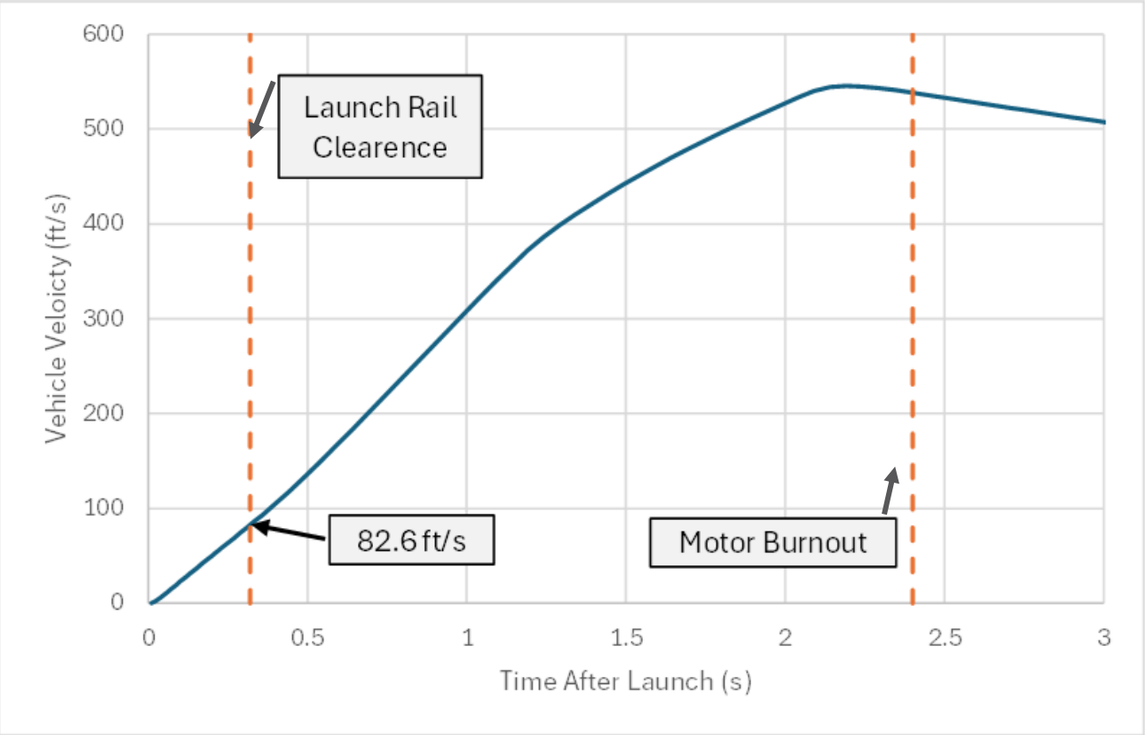


Figure 13 – Flight Velocity Profile

Mass Statement & Mass Margin

Section	Mass (lb)
Payload Section	17.35
Avionics Bay	10.60
Booster Section	17.43
Recovery Section	6.82
Total Dry Mass	≈ 52.20
Total Wet Mass	≈ 57.74

Margin	Kinetic Energy	Descent Time	Apogee
Maximum	54.40 lb	-	63.5 lb
Minimum	-	48.00 lb	35.4 lb

Drogue Recovery Components

- Parachute
  - Type: Hemispherical
  - Diameter = 23 in.
  - Descent Rate = 104 ft/s
- Shock Cord
  - Type: Tubular Kevlar
  - Thickness = 1/2 in.
  - Length = 40 ft.



Figure 14 - Drogue Recovery Components

Main Recovery Components

- Parachute
  - Type: Hemispherical
  - Diameter = 156 in.
  - Descent Rate = 15.3 ft/s
- Recovery Harness
  - Type: Tubular Kevlar
  - Thickness = 1/2 in.
  - Length = 36 ft.



Figure 15 - Main Recovery Components

Recovery CONOPS

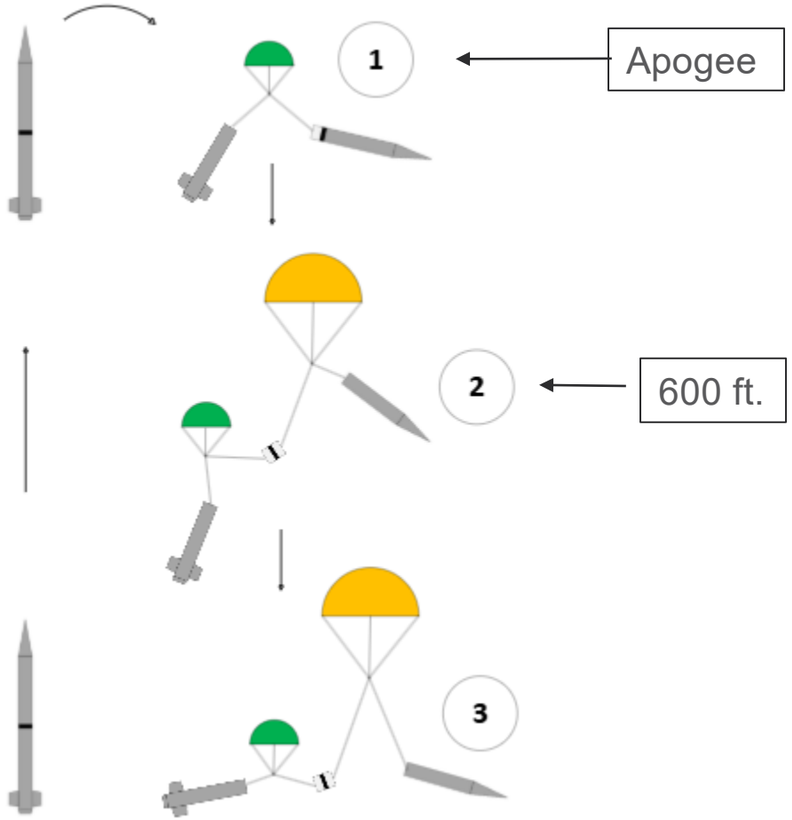


Figure 16 – Recovery Sequence of Events

Vehicle Kinetic Energy

Section	Drogue KE (ft-lbf)	Main KE (ft-lbf)
Payload Section	2,914	63.2
Avionics Bay	1,780	38.6
Booster Section	2,928	63.5

Predicted Drift

Wind Speed (mph)	Drift Distance (ft)
5	569
10	1,138
15	1,706
20	2,275

*Based on vehicle mass with full ballast

Test Plans & Procedures

- Main Parachute Drop Test
- Battery Consumption Test



Figure 17 – Main Parachute Drop Test



Figure 18 – Battery Consumption Test

Subscale Summary

- Scaling factor: 66.7%
- Integrated payload



Figure 19 – Subscale Rocket at the Landing Site



Figure 20 – SSF2 Rocket

Subscale Test Data

- Apogee: 4,287 ft.
- Descent Time: 94.1 s

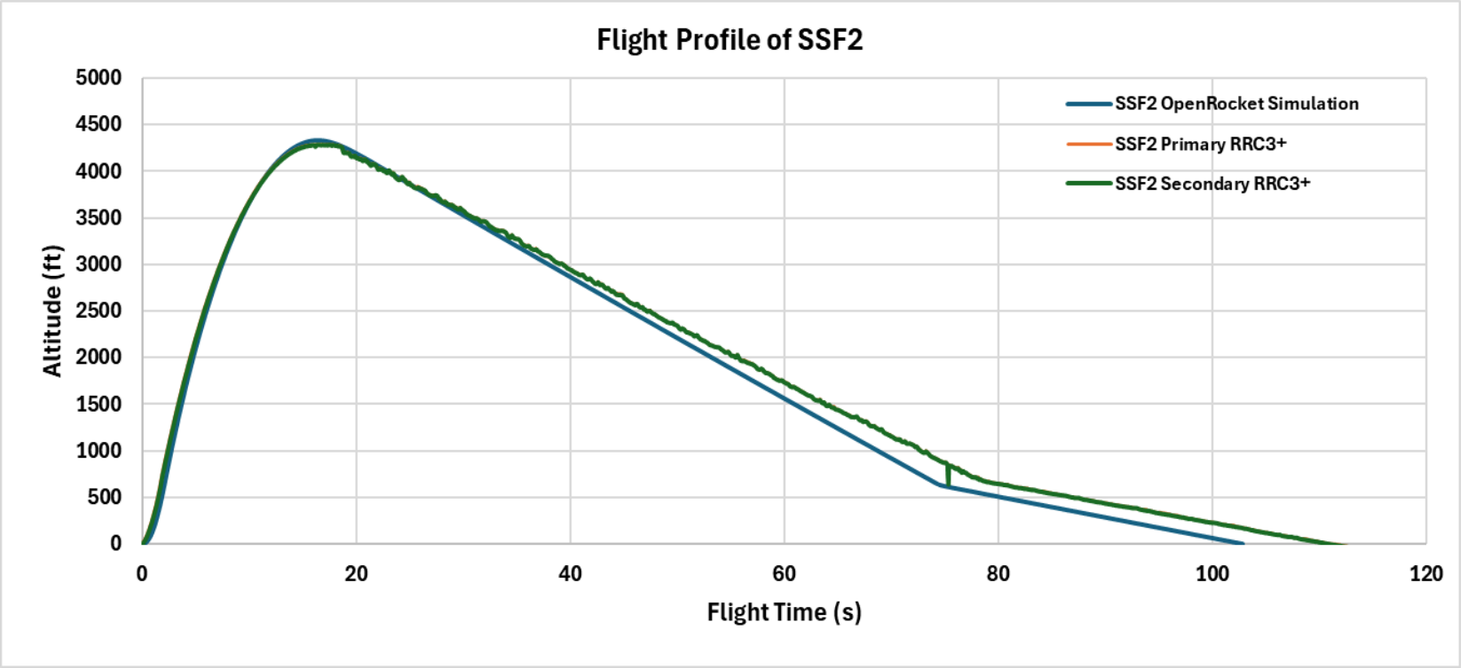


Figure 21 – Plot of Altimeter and OpenRocket Flight Profile

Section	KE (ft-lbf)
Payload Section	41.5
Avionics Bay	21.2
Booster Section	39.4

Tests of Staged Recovery System

- Test Procedure:
  - Test drogue ejection charge (Figure 22)
  - Test main ejection charge (Figure 23)
  - Shock cord should be almost fully extended



Figure 22 - Drogue Ejection Charge Test

Recovery Section	Trial 1 (Initial Calculation)	Trial 2
Drogue	2.2	2.8
Main	2.5	N/A



Figure 23 - Main Ejection Charge Test

Payload Design Review

Payload Design Overview

- Apogee
- Max velocity
- Temperature of landing site
- Time of landing
- 7.04 lbm

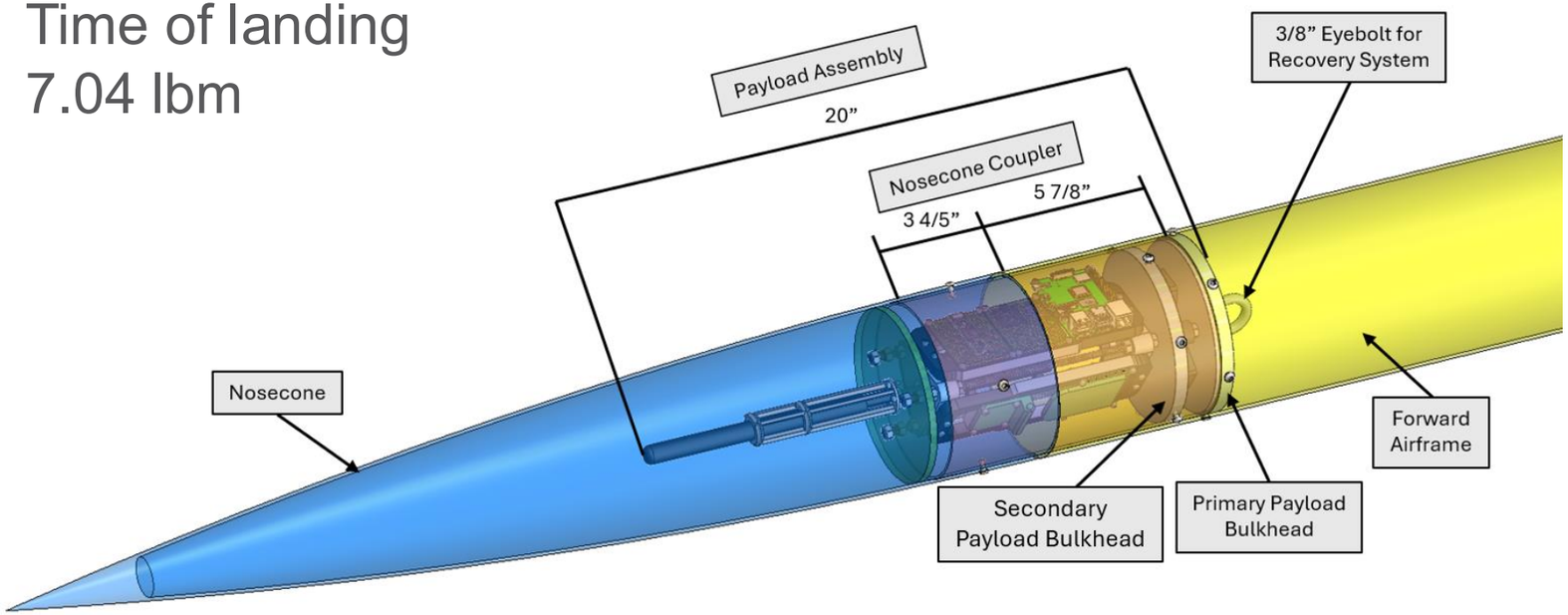


Figure 24 – Payload Design of Integration

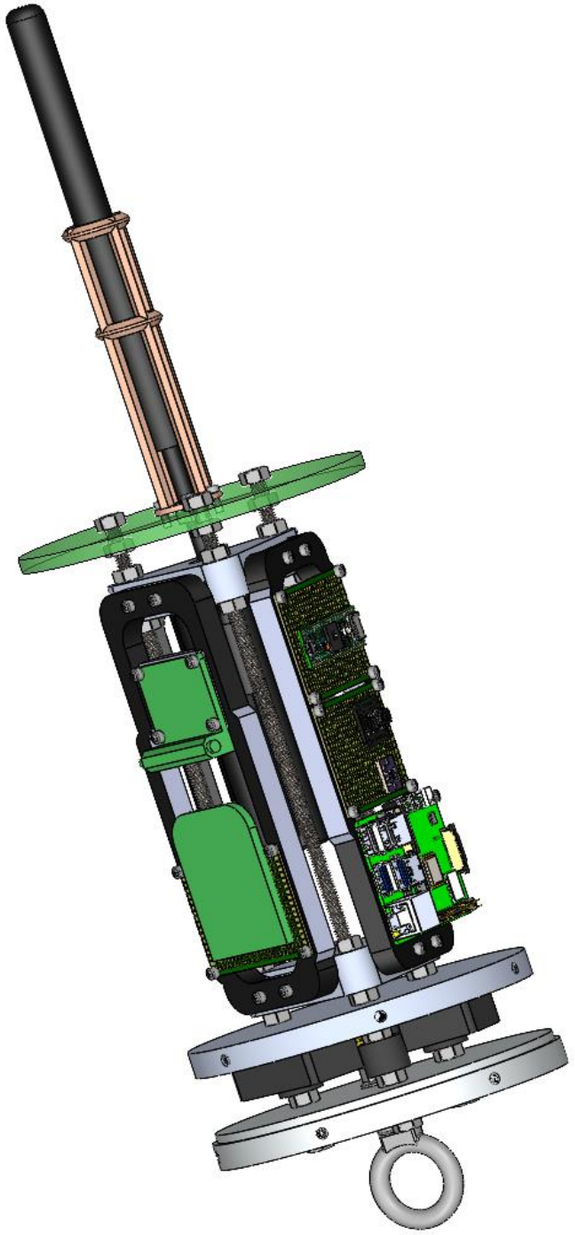


Figure 25 – Payload Capsule Design

Payload Sensors

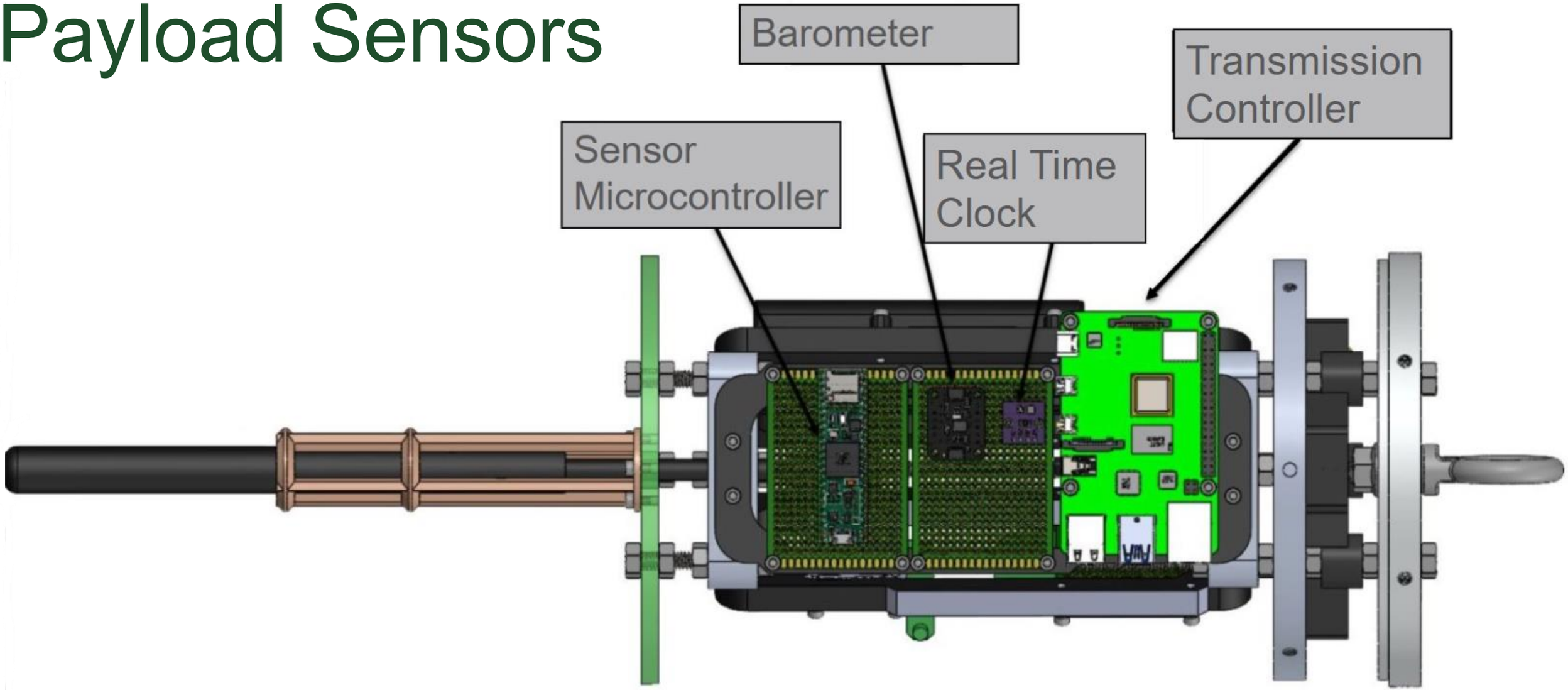


Figure 26 – Payload Sensor Design

Payload Transmission

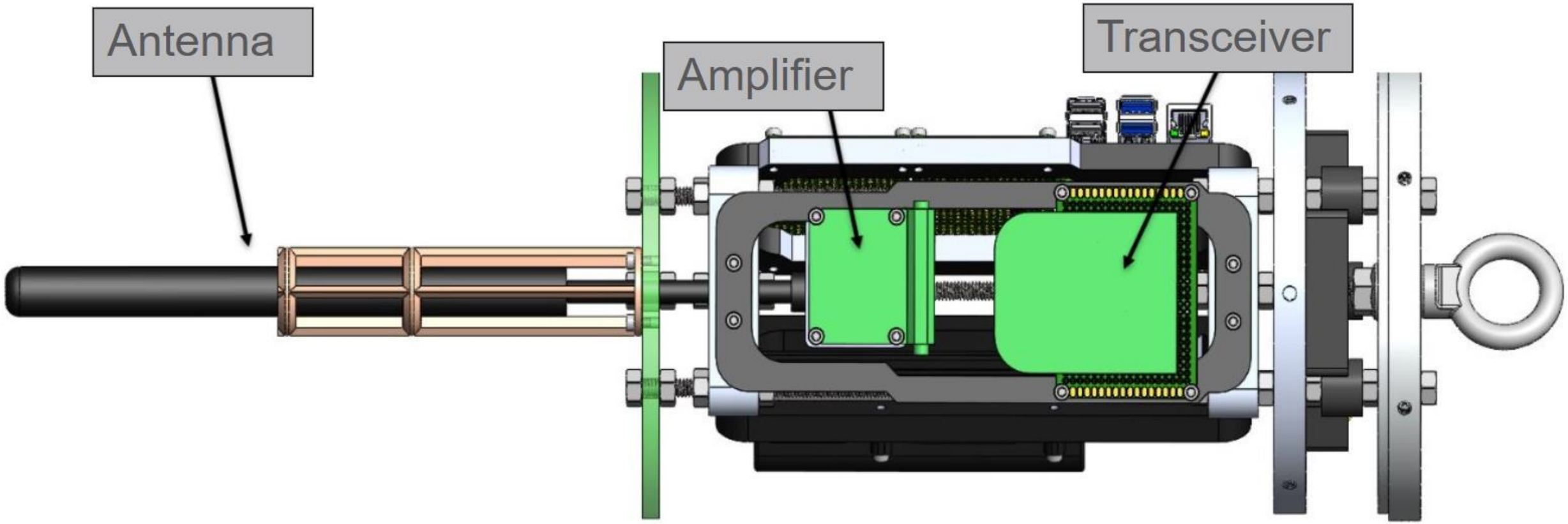


Figure 27 – Payload Transmission Design

Payload Power System

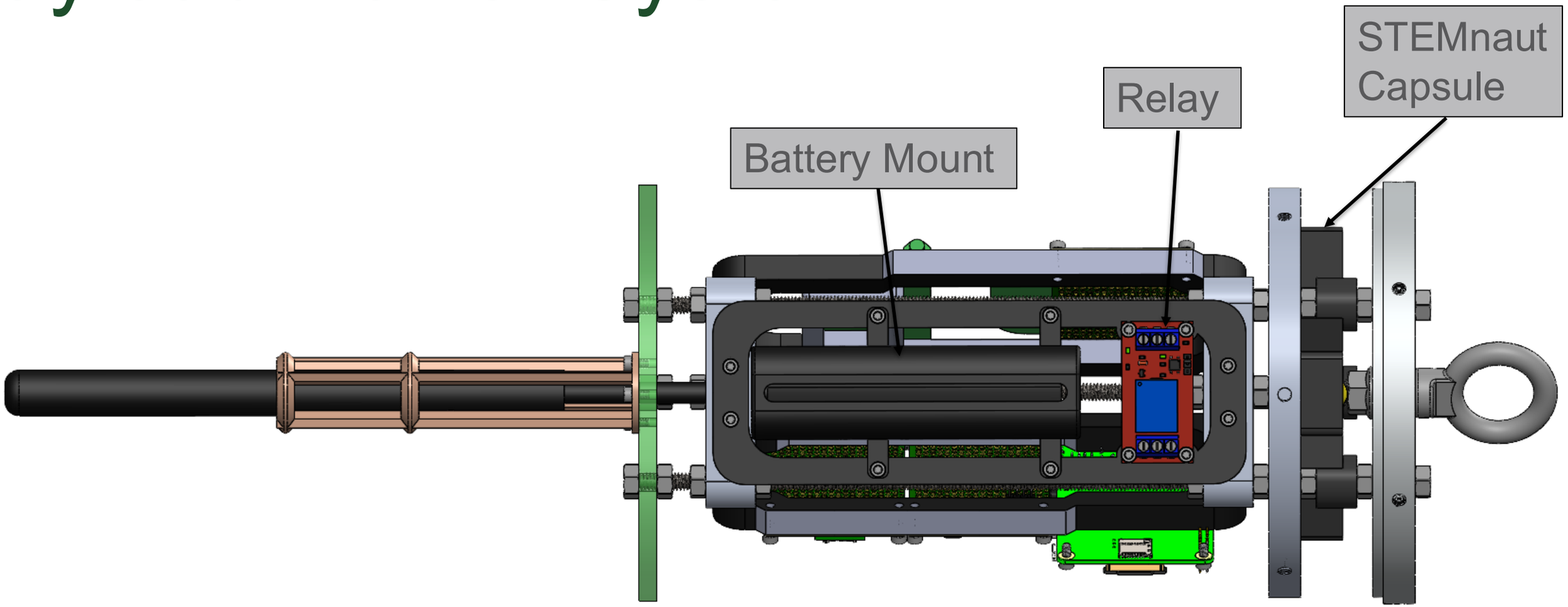


Figure 28 – Payload Power System Design

Payload CONOPS

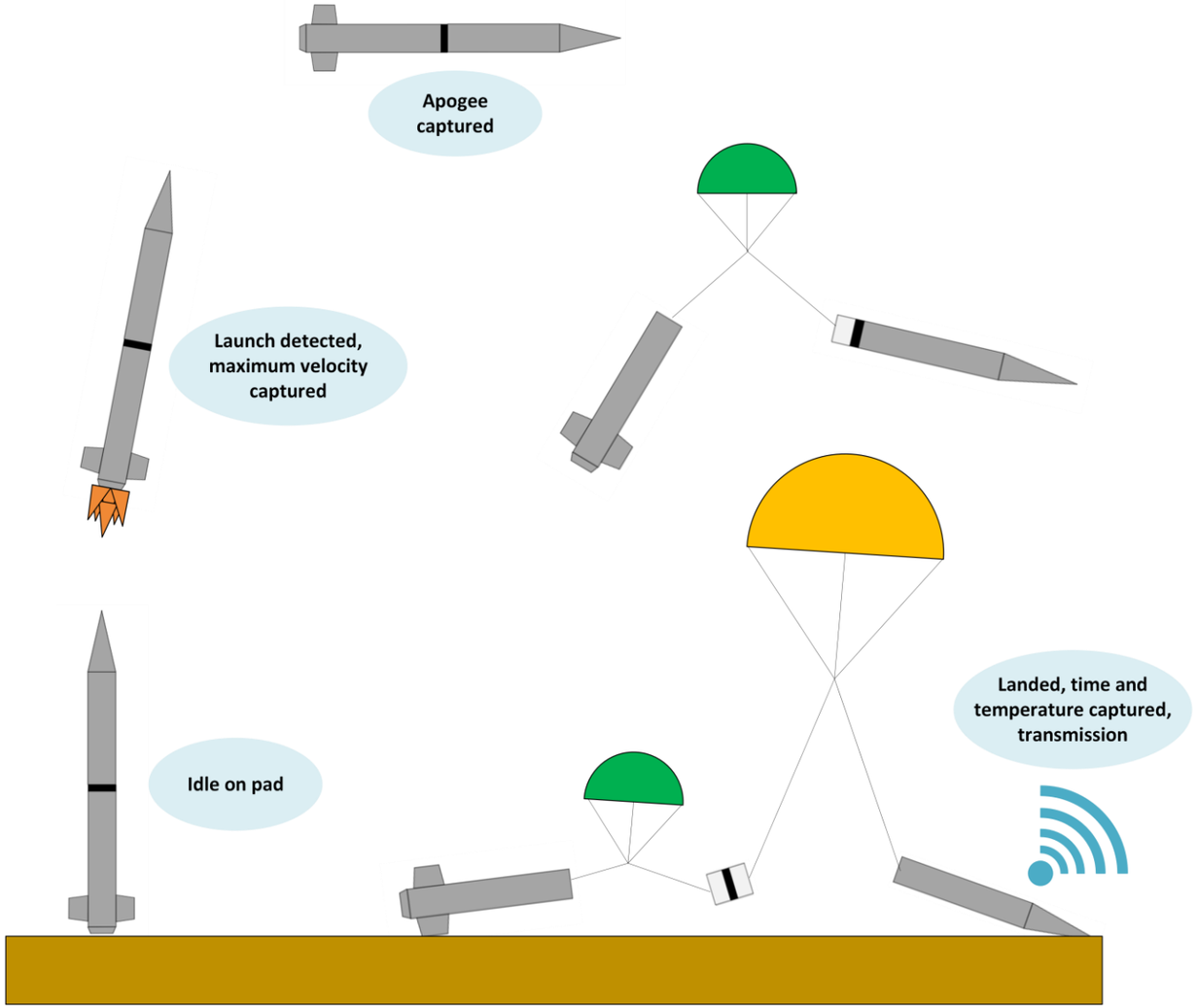


Figure 29 – Payload CONOPS

Flight Capsule



Figure 30 – Flight Capsule Exploded View

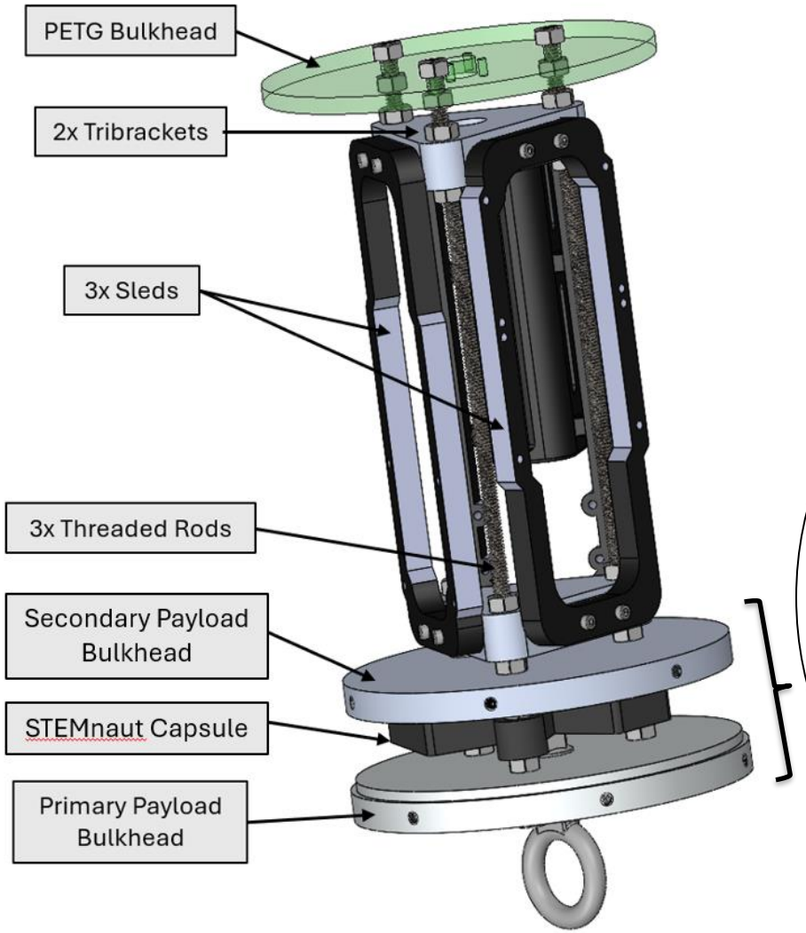


Figure 31 – Flight Capsule Structure

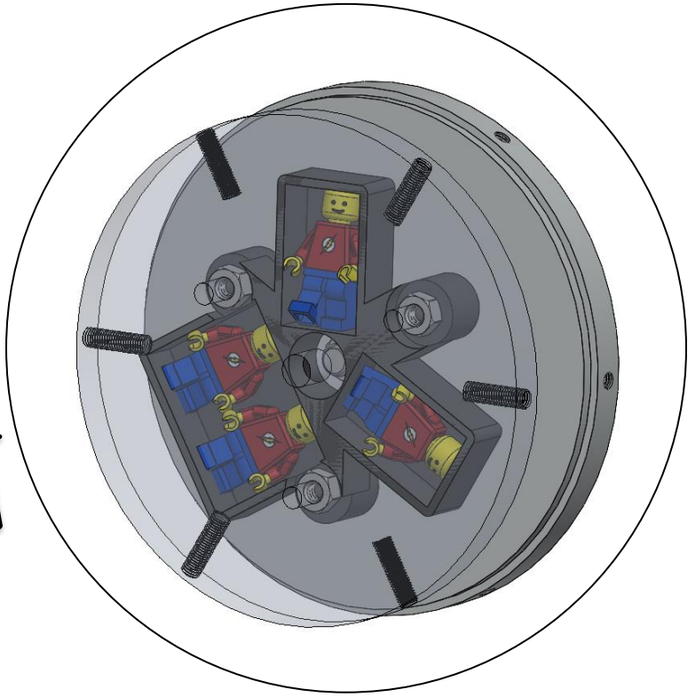


Figure 32 – STEMnaut Capsule

Payload Electrical Design

Sensor	Model	Use
Barometer	BMP384	Apogee & Maximum Velocity
Thermistor	4AT303R44T1A	Temperature
Real Time Clock	PCF8523	Time of Landing

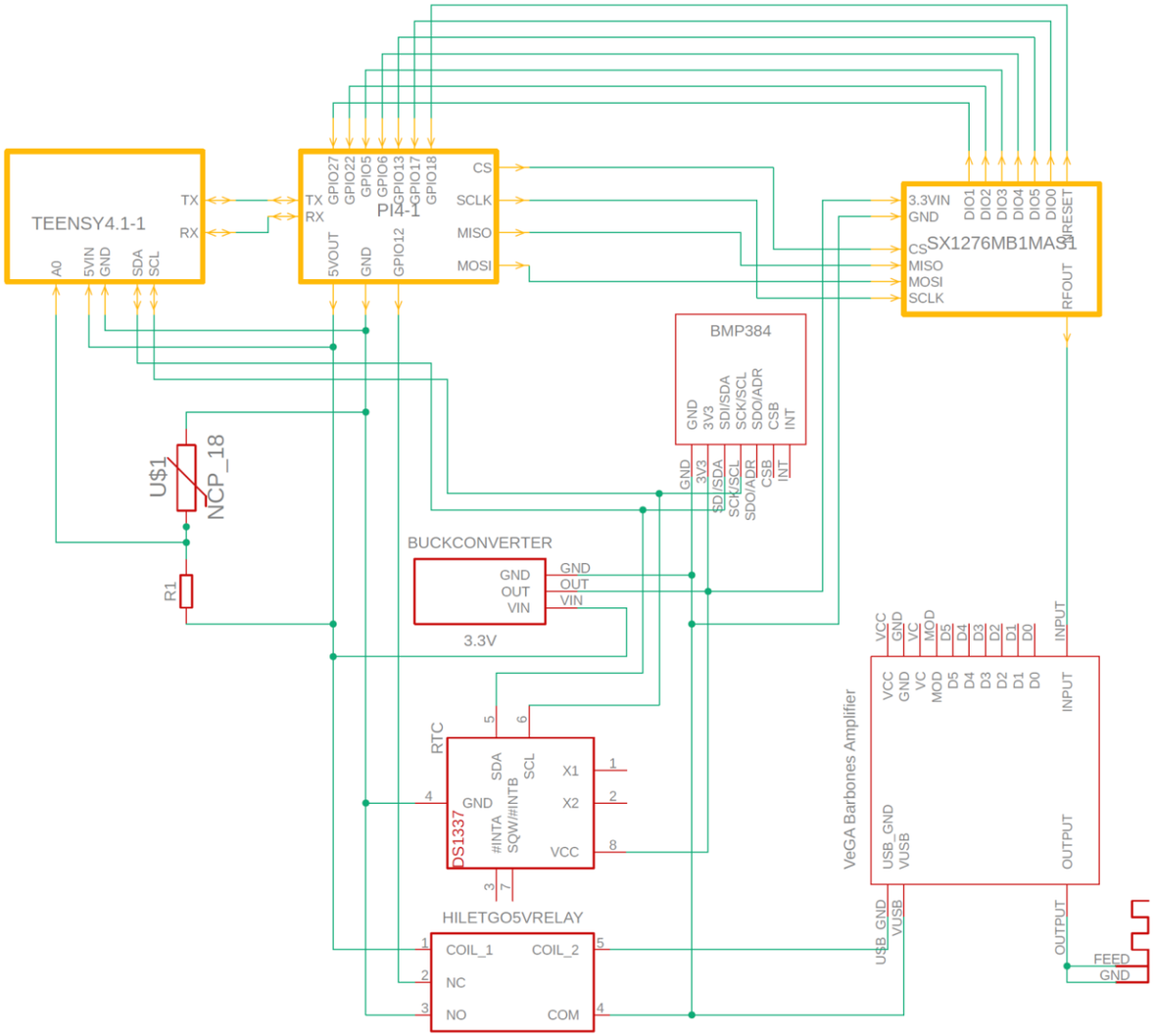


Figure 33 – Payload Wiring Diagram

Transmission Overview

- Relays control power
- Minimum 2500 ft. transmission range

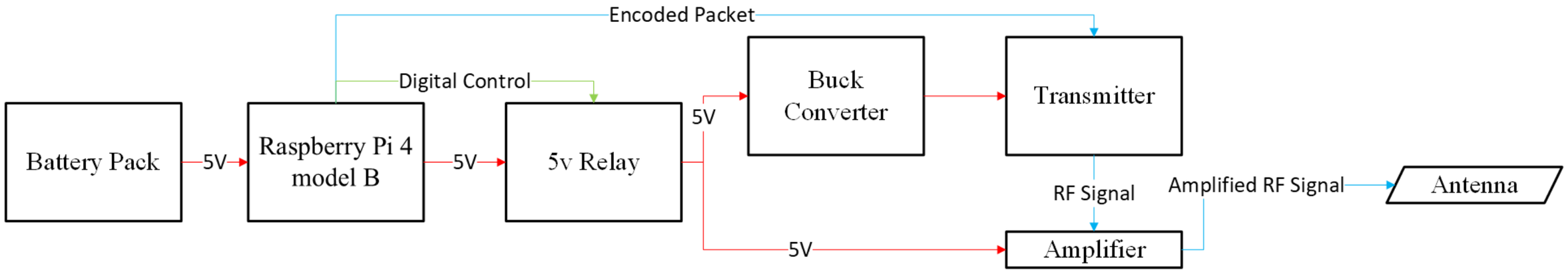


Figure 34 – Transmission Flowchart

Payload Integration Plans

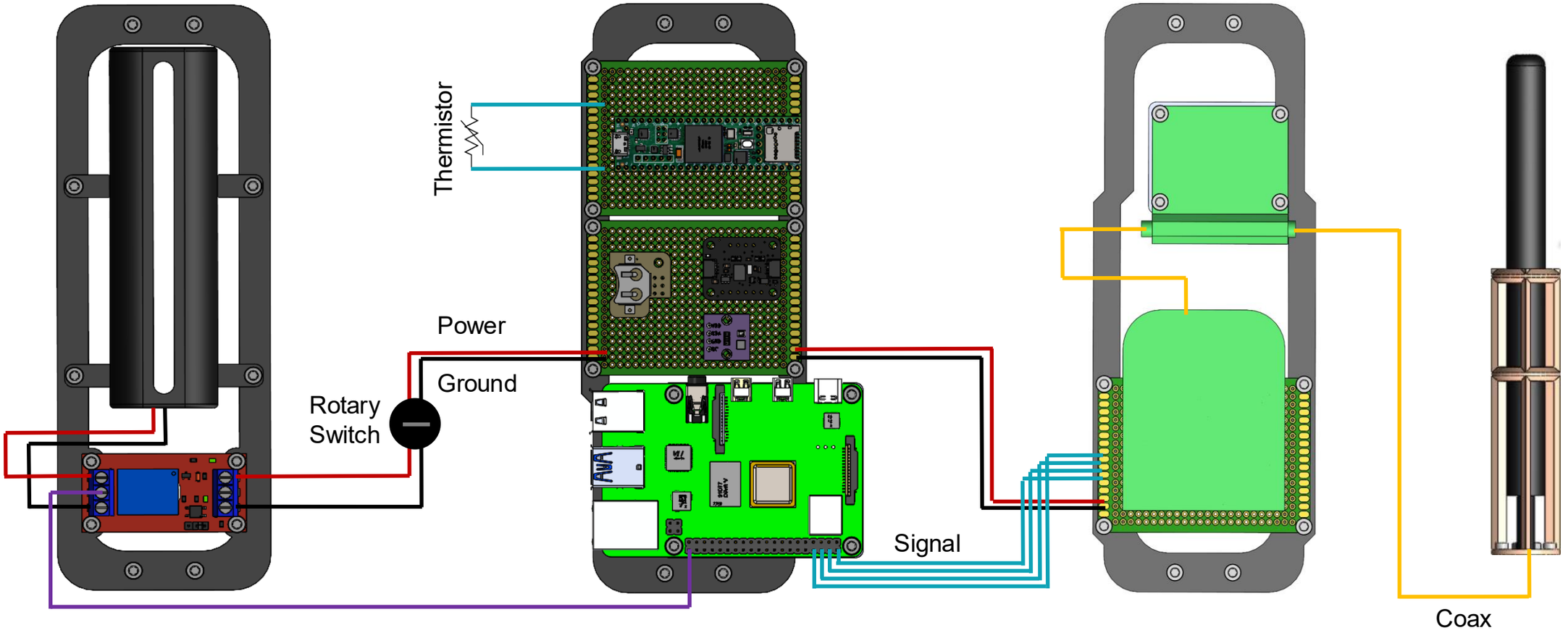


Figure 35 – Payload Integration Plan

Payload-to-Vehicle Interfaces

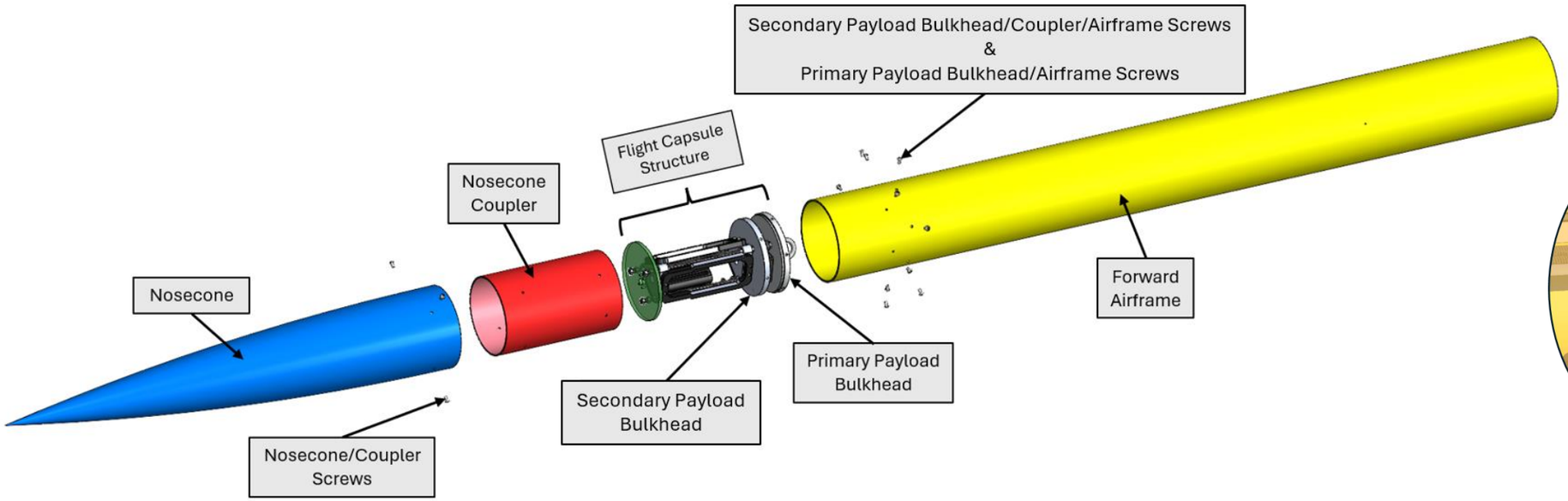


Figure 36 – Payload Integration Exploded View

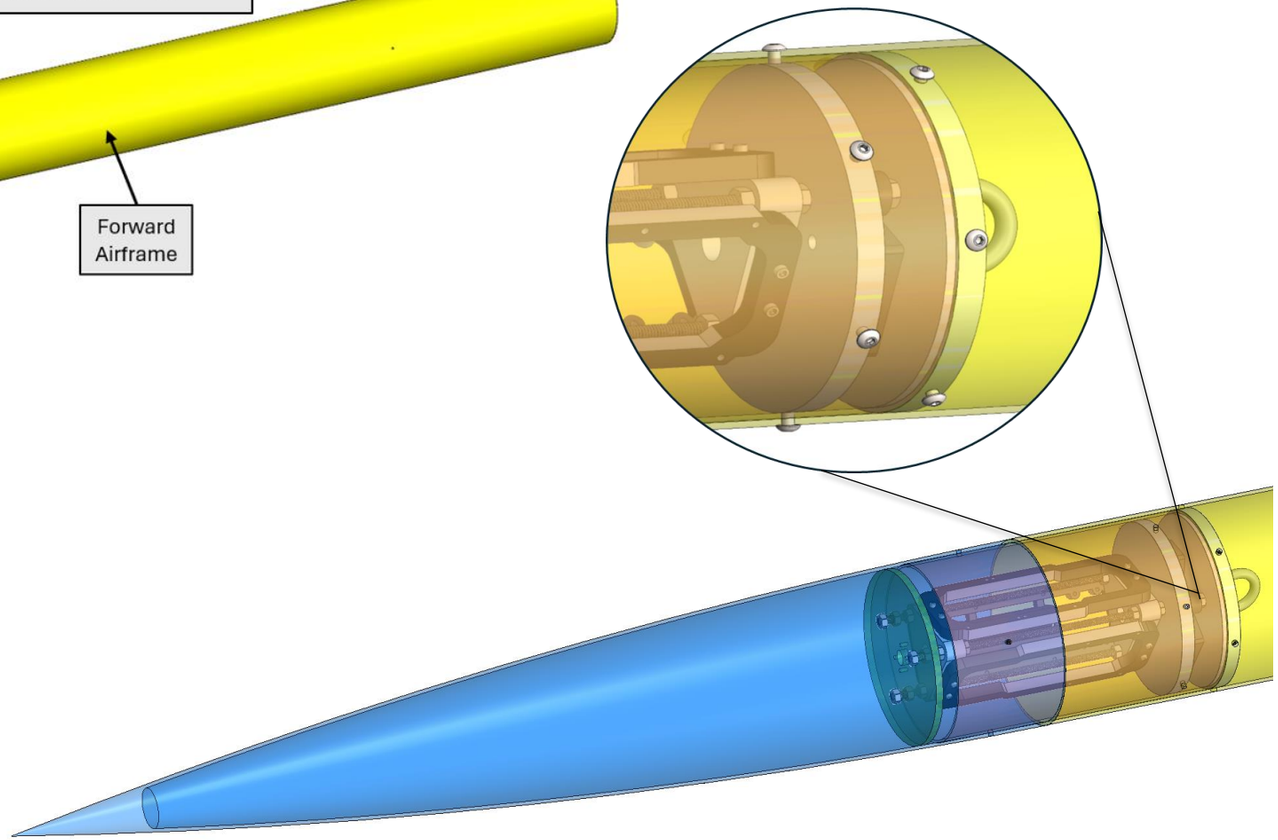


Figure 37 - Interface Screws

Test Plans and Procedures

- Barometer Altitude Accuracy Test
- Transmission Range Test
- Thermistor Response Rate Test

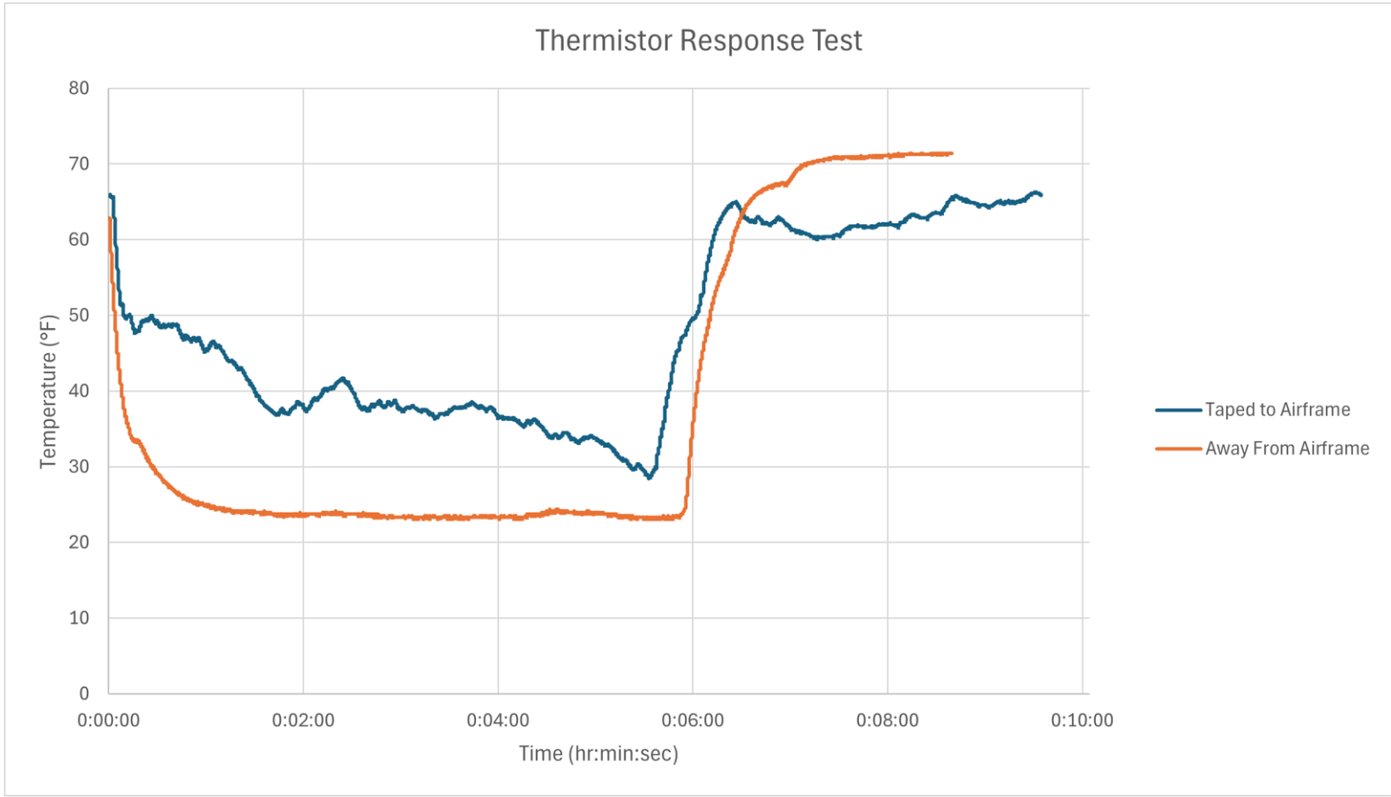


Figure 38 – Thermistor Response Test Data

Project Plan

Project Milestones

Milestone	Date	Status
First Subscale Launch	10/20/24	Completed
Second Subscale Launch	12/7/24	Completed
Full-scale Vehicle Completed	1/26/25	In Progress
Vehicle Demonstration Flight	2/1/25	Start 2/1/25
Payload Completed	1/26/25	In Progress
Payload Demonstration Flight	2/1/25	Start 2/1/25
Competition	5/3/25	Start 5/3/25

Status of Requirements Verification

Requirement Section	In-Progress	Incomplete	Complete	Total
General	1	0	11	12
Vehicle	0	4	18	22
Recovery	1	0	13	14
Payload	2	1	10	13
Safety	0	0	8	8
Final Flight	0	4	1	5


Thank you
Colorado State University